

END TERM EXAMINATION
THIRD SEMESTER [B.TECH] DECEMBER 2017

Paper Code: ETIT-209**Subject: Object Oriented Programming Using C++****Time: 3 Hours****Maximum Marks: 75****Note:** Attempt any five questions including Q. No. 1 which is compulsory.

Q. No.	Question	Marks
1	Answer the following: (a) Explain the features of data hiding and encapsulation in object oriented languages. (b) Mention the limitations of dynamic binding. (c) Give the applications of static functions in a class. (d) Give the syntax of copy constructor and explain its working. (e) Compare the features of class and structure. (f) Explain namespace. Give an example in C++. (g) Explain how Unexpected Exceptions are processed in C++. (h) Give the syntax and explain: Write a statement that opens file trans.dat for input using an ifstream object called inTransaction. (i) Explain iterators in STL. (j) Explain pure virtual functions. Give its syntax.	10 × 2.5 = 25
2	(a) Compare object oriented programming and structured programming. (5 Marks) (b) Write a C++ program to illustrate new and delete operators. (5 Marks) (c) Mention the role of indirection operator (*) and address operator (&) in C++. (2.5 Marks)	5 + 5 + 2.5 = 12.5
3	(a) Define function overloading. Write a C++ program to swap two numbers of type integers, float and character using function overloading. (6.5 Marks) (b) Explain various types of constructors. Give an example for each. (6 Marks)	6.5 + 6 = 12.5
4	(a) Explain the sequence of constructors and destructors invoked for multiple inheritance. Give an example for illustration. (4 Marks) (b) Write a C++ program to illustrate virtual functions. (6 Marks) (c) Justify the false statement and give its solution: 'A class is made abstract by declaring that class virtual.' (2.5 Marks)	4 + 6 + 2.5 = 12.5

[P.T.O.]

Q. No.	Question	Marks
5	(a) Write a C++ program to illustrate class templates. (5 Marks) (b) Compare sequence containers and associative containers in STL. Give an example for each. (5 Marks) (c) Mention various types of Iterator Categories in STL. (2.5 Marks)	5 + 5 + 2.5 = 12.5
6	(a) Find the error and show how to correct the following statement: (3.5 Marks) <i>'The file "tools.dat" should be opened to add data to file without discarding the current data.'</i> <code>ofstream outTools("tools.dat", ios::out);</code> (b) Justify the statement: <i>'Each automatic object constructed in a try block is destructed before an exception is thrown'</i>. Give an example. (4 Marks) (c) Write a C++ program to create a User-defined Exception Class DivisionByZero. In main() user enters dividend and divisor. If divisor is 0, control passes to user-defined exception class to display <i>'Division by zero! Invalid.'</i> (5 Marks)	3.5 + 4 + 5 = 12.5
7	(a) Compare inheritance and composition. Give their respective applications. (4 Marks) (b) Write a C++ program to overload function templates. (4.5 Marks) (c) Give an example in C++ to illustrate Rethrowing an Exception. (4 Marks)	4 + 4.5 + 4 = 12.5
8	Write short notes on the following: (3 × 4.16 ≈ 12.5 Marks) (a) Standard Template Library (STL) (b) Generic Programming in C++ (c) Fundamental Features of Object-Oriented Programming	12.5
